

ORF245 Final Exam Solutions

Fall 2023

Part I

1. False. E.g., could just have $P(A) = 0$. But can also have $P(A|B) = 0$ even if $P(A) > 0$.

2. True. This is what is often called the union bound.

$$3. \quad \frac{1}{2^{10}} = \frac{1}{1024}$$

$$4. \quad \binom{52}{5} = \frac{52!}{5! 47!}$$

5. True.

6. True.

7. False. This does not hold for any pdf. A similar property does hold for the cdf.

8. False. $F_X(x) \rightarrow 1$ as $x \rightarrow \infty$, $F_Y(y) \rightarrow 1$ as $y \rightarrow \infty$ and $F_{X+Y}(u) \rightarrow 1$ as $u \rightarrow \infty$, so the property can't be true.

$$\begin{aligned} 9. \quad \text{True.} \quad \text{Cov}(2X, Y) &= E[(2X - 2\bar{X})(Y - \bar{Y})] \\ &= 2 E[(X - \bar{X})(Y - \bar{Y})] \\ &= 2 \text{Cov}(X, Y) \end{aligned}$$

10. False. Many examples will show this. For example, consider $X=1$ and $X=-1$ each with probability $1/2$ and let $g(x) = x^2$. $g(E[X]) = 0$, $E[g(X)] = 1$
11. True. Basic property of expectation.
12. False. Only approximately Gaussian.
13. $P(X=\mu) = 0$. The probability that X is exactly equal to any particular value is 0.
14. False. Point estimation gives a single value as an estimate. Confidence intervals give a range of values.
15. False. $\hat{\theta}$ is unbiased if $E[\theta - \hat{\theta}] = 0$, i.e. without the square.

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Part I

16- False. A common example is for λ if $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$,
or for σ^2 in $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where $\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$.

17- The Interval is the only random part.

18- False - we can have upper/lower / two-sided $1-\alpha$ CI's

19- True - the intervals are random, so they can lie far apart

20- False - CI width is $2 \cdot z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$, doubling n reduces width by
a factor of $\sqrt{2}$.

21- False

22- True

23- True

24- False

25- False - the opposite is true

26- True - since t_{10} has more mass on the tails

$$27. \underline{r^2 \in [0,1]}$$

$$28. \underline{T_{true}}$$

$$29. \underline{T_{true}}$$

$$30. \hat{\alpha}_1 = \frac{\hat{\text{Cov}}(Z, Y)}{\hat{\text{Var}}(Z)} = 2 \cdot \frac{\hat{\text{Cov}}(X, Y)}{4 \cdot \hat{\text{Var}}(X)} = \frac{1}{2} \cdot \hat{\beta}_1$$

Part II

1. (a) (1 point)

Let 1_B = event that marble 1 is blue.

$$P(1_B) = \frac{3}{6} = \frac{1}{2}$$

(b) (2 points)

Let 2_B = event that marble 2 is blue.

$$P(1_B \text{ and } 2_B) = P(1_B)P(2_B|1_B) = \left(\frac{3}{6}\right)\left(\frac{2}{5}\right) = \frac{1}{5}$$

(c) (2 points)

Let $1_N, 2_N, 3_N$ be the events that marbles 1, 2, 3 are not blue, respectively.

$$\begin{aligned} P(1_N \text{ and } 2_N \text{ and } 3_N) &= P(1_N)P(2_N|1_N)P(3_N|1_N, 2_N) \\ &= \left(\frac{3}{6}\right)\left(\frac{2}{5}\right)\left(\frac{1}{4}\right) = \frac{1}{20} \end{aligned}$$

(d) (5 points)

Let X = # of draws needed to get blue marble.

$$P(X=1) = \frac{1}{2} \quad P(X=2) = \left(\frac{1}{2}\right)\left(\frac{3}{5}\right) = \frac{3}{10}$$

$$P(X=3) = \left(\frac{1}{2}\right)\left(\frac{2}{5}\right)\left(\frac{3}{4}\right) = \frac{3}{20} \quad P(X=4) = \left(\frac{1}{2}\right)\left(\frac{2}{5}\right)\left(\frac{1}{4}\right)(1) = \frac{1}{20}$$

$$E[X] = \left(\frac{1}{2}\right)(1) + \left(\frac{3}{10}\right)(2) + \left(\frac{3}{20}\right)(3) + \left(\frac{1}{20}\right)(4) = \frac{35}{20} = \frac{7}{4} = 1.75$$

2. The given pdf has the form of a normal/Gaussian pdf

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(a) (2 points)

By inspection, $\mu = E[X] = -\pi$

and $2\sigma^2 = 1$ so $\sigma^2 = V[X] = \frac{1}{2}$

(b) (2 points)

$$E[X^2] = V[X] + (E[X])^2 = \frac{1}{2} + (-\pi)^2 = \frac{1}{2} + \pi^2$$

(c) (3 points)

$$Y = 2 - \frac{X}{\pi}$$

$$E[Y] = E\left[2 - \frac{X}{\pi}\right] = 2 - \frac{E[X]}{\pi} = 2 + \frac{\pi}{\pi} = 3$$

$$V[Y] = V\left[2 - \frac{X}{\pi}\right] = V[2] + V\left[\frac{X}{\pi}\right]$$

$$= 0 + \frac{1}{\pi^2} V[X] = \frac{1}{2\pi^2}$$

(d) (3 points)

$$P[X > \pi] = P[X + \pi > 2\pi] = P\left[\frac{X + \pi}{(\frac{1}{\sqrt{2}})} > 2\sqrt{2}\pi\right]$$

$$= 1 - \Phi(2\sqrt{2}\pi)$$

$$3. \quad P(X=k) = \begin{cases} \frac{e^{-\lambda} \lambda^k}{k!} & k=0,1,2,\dots \\ 0 & \text{otherwise} \end{cases}$$

(a) (2 points)

$$L(\lambda; 8) = \frac{e^{-\lambda} \lambda^8}{8!}$$

$$\log L(\lambda; 8) = -\lambda + 8 \ln \lambda - \ln 8!$$

(b) (2 points)

$$\frac{\partial \log L(\lambda; 8)}{\partial \lambda} = -1 + \frac{8}{\lambda}$$

$$\frac{\partial \log L(\lambda; 8)}{\partial \lambda} = 0 \Rightarrow -1 + \frac{8}{\hat{\lambda}_{MLE}} = 0 \quad \text{so } \hat{\lambda}_{MLE} = 8$$

(c) (2 points)

$$L(\lambda; x_1, \dots, x_n) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{x_1! \cdots x_n!}$$

$$\log L(\lambda; x_1, \dots, x_n) = -n\lambda + \left(\sum_{i=1}^n x_i\right) \ln \lambda + \ln(x_1! \cdots x_n!)$$

(d) (2 points)

$$\frac{\partial \log L(\lambda; x_1, \dots, x_n)}{\partial \lambda} = 0 \Rightarrow -n + \frac{1}{\lambda} \left(\sum_{i=1}^n x_i\right) = 0$$

$$\text{so } \hat{\lambda}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

(e) (2 points)

Unbiased. $E[\hat{\lambda}_{MLE}] = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{1}{n} \sum_{i=1}^n E[x_i]$
 $= \frac{1}{n} \sum_{i=1}^n \lambda = \lambda.$

Part II

4a- $\hat{\theta}_{Avg} = 2 \cdot \bar{X}$

Take $E[\cdot]$

$$E[\hat{\theta}_{Avg}] = 2 \cdot E[\bar{X}] = 2 \cdot E\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

$$= 2 \cdot \frac{1}{n} \sum_{i=1}^n E[X_i]$$

$$= 2 \cdot \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{2}\theta\right)$$

$$E[\hat{\theta}_{Avg}] = \theta$$

$\Rightarrow$ Unbiased!

$$E[X_i] = \int_0^{\theta} \frac{1}{\theta} \cdot x \, dx$$

$$= \frac{1}{\theta} \cdot \left[\frac{1}{2} \theta^2 - \frac{1}{2} 0^2 \right] = \frac{1}{2} \theta$$

4b- $Var(\hat{\theta}_{Avg}) = Var(2 \cdot \bar{X}) = 4 \cdot Var(\bar{X})$

$$= 4 \cdot Var\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$$

$$= \frac{4}{n^2} \cdot Var\left(\sum_{i=1}^n X_i\right)$$

$$= \frac{4}{n^2} \cdot \sum_{i=1}^n Var(X_i) \quad \left\{ \text{By } X_i \text{'s iid} \right.$$

$$E[X_i^2] = \int_0^{\theta} \frac{1}{\theta} x^2 \, dx = \frac{1}{\theta} \cdot \left[\frac{1}{3} \theta^3 - \frac{1}{3} \cdot 0 \right] = \frac{1}{3} \theta^2$$

$$Var(X_i) = E[X_i^2] - E[X_i]^2$$

$$= \frac{1}{3} \theta^2 - \left(\frac{1}{2} \theta\right)^2 = \frac{1}{12} \theta^2$$

$$Var(\hat{\theta}_{Avg}) = \frac{4}{n^2} \cdot \sum_{i=1}^n \frac{1}{12} \theta^2 = \frac{1}{3} \cdot \frac{\theta^2}{n}$$

4c- $E[\hat{\theta}_{MLE}] = E[X_{(n)}]$

pdf of $X_{(n)}$ is $\frac{d}{dx} F_{(n)}(x) = \frac{n}{\theta} \cdot x^{n-1}$

$$= \int_0^{\theta} \frac{n}{\theta} x^{n-1} \cdot x \cdot dx$$

$$= \frac{n}{\theta} \cdot \int_0^{\theta} x^n \, dx = \frac{n}{\theta} \left[\frac{1}{n+1} \theta^{n+1} - \frac{1}{n+1} 0^{n+1} \right]$$

$$E[\hat{\theta}_{MLE}] = \frac{n}{n+1} \cdot \theta$$

$\Rightarrow$ Biased!

4d- First, compute $E[\hat{\theta}_{MLE}^2] = \int_0^\theta \frac{n}{\theta^n} x^{n-1} \cdot x^2 \cdot dx$

$$= \frac{n}{\theta^n} \left[\frac{1}{n+2} \cdot \theta^{n+2} - \frac{1}{n+2} \cdot 0^{n+2} \right]$$

$$E[\hat{\theta}_{MLE}^2] = \frac{n}{n+2} \cdot \theta^2$$

$$So: Var(\hat{\theta}_{MLE}) = E[\hat{\theta}_{MLE}^2] - E[\hat{\theta}_{MLE}]^2$$

$$= \frac{n}{n+2} \cdot \theta^2 - \left(\frac{n}{n+1} \theta \right)^2$$

$$= \theta^2 \left[\frac{n}{n+2} - \frac{n^2}{(n+1)^2} \right]$$

$$= \frac{n(n+1)^2}{(n+2)(n+1)^2} - \frac{n^2(n+2)}{(n+2)(n+1)^2} = \frac{n(\cancel{n+2}n+1 - \cancel{n^2} - 2n)}{(n+2)(n+1)^2}$$

$$\underline{Var(\hat{\theta}_{MLE}) = \theta^2 \cdot \frac{n}{(n+2)(n+1)^2}}$$

5a- Since we have 81 samples, the CLT is a good approx.

$$\bar{X} \overset{\text{approx}}{\sim} N(\mu_1, \frac{s_1^2}{81})$$

$$\bar{Y} \overset{\text{approx}}{\sim} N(\mu_2, \frac{s_2^2}{81})$$

Hence, $\underline{\bar{X} - \bar{Y}} \overset{\text{approx}}{\sim} N(\mu_1 - \mu_2, \underline{\frac{s_1^2 + s_2^2}{81}})$

$$s_1^2 + s_2^2 = 0.9^2 + 1.2^2 = 1.5^2$$

5b- From Before, $\bar{X} - \bar{Y} \overset{\text{approx}}{\sim} N(\mu_b - \mu_a, (\frac{1.5}{q})^2)$

$$\Rightarrow \frac{\bar{X} - \bar{Y} - \mu_\Delta}{1.5/q} \sim N(0, 1)$$

So: $[\bar{X} - \bar{Y} - z_\alpha \cdot \frac{1.5}{q}, \infty)$ will give a lower $1-\alpha$ CI for μ_Δ

Plugging in: $[10.6 - 9.5 - 1.645 \cdot \frac{1.5}{q}, \infty)$

$$\Leftrightarrow \underline{\underline{[1.1 - 1/6 \cdot 1.645, \infty)}}$$

5c- Compute the value of $\frac{\bar{X} - \bar{Y} - \frac{1}{2}}{1.5/q}$ From μ_Δ under H_0

$$\frac{10.6 - 9.5 - 0.5}{1/6} = \frac{0.6}{1/6} = \underline{\underline{3.6}}$$

↑ This is bigger than $z_{5\%}$, so its p-value will be less than 5%.

Hence, we reject H_0 .

5d- We do not reject H_0 when $\frac{\bar{X} - \bar{Y} - 0.5}{1/6} \leq z_\alpha = 1.645$

But if $\mu_\Delta = 1$, then:

$$\frac{\bar{X} - \bar{Y} - 0.5}{1/6} = \underbrace{\frac{\bar{X} - \bar{Y} - 1}{1/6}}_{\sim N(0, 1)} + \frac{0.5}{1/6}$$

$$\text{So: } \frac{\bar{X} - \bar{Y} - 0.5}{1/6} \sim N(2, 1)$$

$$\begin{aligned} \text{Hence, } P(\text{Type II Error}) &= P\left(\frac{\bar{X} - \bar{Y} - 0.5}{1/6} \leq z_\alpha \mid \mu_\Delta = 1\right) \\ &= P\left(\frac{\bar{X} - \bar{Y} - 1}{1/6} \leq z_\alpha - 3 \mid \mu_\Delta = 1\right) \\ &= \Phi(z_\alpha - 3) \end{aligned}$$

$$\underline{\underline{P(\text{Type II Error}) = \Phi(1.645 - 3)}}$$

$$6a - \text{Compute: } \widehat{Cov}(x, y) = \frac{1}{100} \sum_{i=1}^{100} x_i y_i - \left(\frac{1}{100} \sum_{i=1}^{100} x_i \right) \left(\frac{1}{100} \sum_{i=1}^{100} y_i \right)$$

$$= 15 - 3 \cdot 4 = \underline{\underline{3}}$$

$$\widehat{Var}(x) = \frac{1}{100} \sum_{i=1}^{100} x_i^2 - \left(\frac{1}{100} \sum_{i=1}^{100} x_i \right)^2$$

$$= 17 - 3^2 = \underline{\underline{4}}$$

$$\text{So: } \underline{\underline{\hat{\beta}_1 = \frac{\widehat{Cov}(x, y)}{\widehat{Var}(x)} = \frac{3}{4}}}$$

$$6b - SE(\hat{\beta}_1) = \frac{\hat{\sigma}_\varepsilon}{\sqrt{n \cdot \widehat{Var}(x)}} = \frac{10}{10 \cdot \sqrt{4}} = \underline{\underline{\frac{1}{2}}}$$

$$6c - \underline{\text{No.}} \quad \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = 1.5, \text{ which is less (in absolute value) than } z_{2.5\%} = 1.96.$$

6d - After quadrupling the data, neither $\widehat{Cov}(x, y)$ nor $\widehat{Var}(x)$ will change. So, we still have $\underline{\underline{\hat{\beta}_1 = \frac{3}{4}}}$

$$6e - \text{Now, } SE(\hat{\beta}_1) = \frac{\hat{\sigma}_\varepsilon}{\sqrt{4 \cdot 100 \cdot \widehat{Var}(x)}} = \frac{1}{2} \cdot \frac{10}{10 \cdot \sqrt{4}} = \frac{1}{2} \cdot \frac{1}{2} = \underline{\underline{\frac{1}{4}}}$$

$$\text{So, } \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = \frac{3/4}{1/4} = \underline{\underline{3}}. \text{ Since } |3| > 1.96, \text{ we } \underline{\underline{\text{reject the null.}}}$$

6f - $\hat{\sigma}_\varepsilon^2 = \frac{SSE}{n-2}$. Quadrupling the data will quadruple SSE (since our estimates don't change), but $n-2$ goes from 98 to 398, which is not quadrupled. Hence, we should have

$$\hat{\sigma}_\varepsilon^2 = \frac{4 \cdot (100 \cdot 98)}{398}, \text{ since SSE is originally } 100 \cdot 98, \text{ and}$$

$$\text{hence } \underline{\underline{\hat{\sigma}_\varepsilon^2 = 100 \cdot \frac{4 \cdot 98}{398} \neq 100.}}$$